



Contributing to open source as it was intended, improving the code and publishing it for others, or hiring professionals to manage the OSS code and debug it as necessary are options. However, the industry cannot afford to not contribute anything at all.

HFCL introduces open source Wi-Fi 7 access points

The world's first open source Wi-Fi 7 access points were introduced by HFCL Limited, in collaboration with Qualcomm Technologies, Inc., on October 1, 2022, at the India Mobile Congress in Pragati Maidan, New Delhi.

The access points are based on IEEE 802.11be, a ground-breaking Wi-Fi technology that is intended to give extremely high throughput (EHT), increased spectrum efficiency, better interference mitigation, and support for real-time applications (RTA). In order to provide a better user experience while using less power, Wi-Fi 7 uses faster connections with 320MHz and 4kQAM, numerous connections with multi-link operation, and adaptive connections for adaptive interference puncturing.



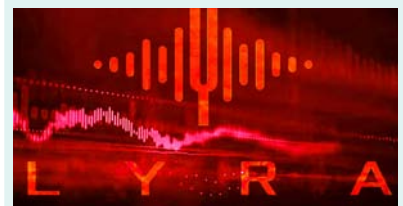
Wi-Fi 7 promises a significant technological advance over all prior Wi-Fi standards, providing a more immersive user experience and paving the way for a more robust digital future. The peak data speeds supported by HFCL's Wi-Fi 7 APs will exceed 10Gbps, and they will have latency under 2ms as opposed to the 5Gbps and 10ms of existing Wi-Fi 6 products.

Lyra audio codec gets an update

The open source Lyra audio codec, which Google first released last year, has been updated with Lyra V2, which has a new architecture that enables scalable bitrate capabilities, greater performance, higher audio quality, and compatibility with more systems.

Lyra V2's internal architecture is based on the SoundStream end-to-end neural audio codec, which has a 'residual vector quantizer' (RVQ) sitting before and after the transmission channel and the ability to adjust the amount of quantizers to be used at any time to alter the audio bitrate. The three available bitrates are 3.2kbps, 6kbps, and 9.2kbps. Lyra V2 makes use of artificial intelligence, and a TensorFlow Lite model makes it possible for it to function on Linux and Android smartphones. Support for Mac and Windows is experimental. At this time, iOS and other embedded platforms are not supported; however, things might change later.

As compared to other audio codecs like Lyra (V1) and Opus, the new audio codec offers a higher quality (MUSHRA score) than those at a given bitrate. Lyra V2 at 9.2kbps offers roughly the same quality as Opus at 14kbps.



This second generation codec's latency has improved from 100ms to 20ms, making it similar to Opus for WebRTC, which typically has delays of 26.5ms, 46.5ms, and 66.5ms. In order to provide real-time audio encoding/decoding and reduce power consumption, Lyra V2 also encodes and decodes five times quicker than Lyra V1.